



BRIDGE ENGINEERING CHALLENGE

What is a Bridge?



Meera's Big Problem!

Meera needs to visit her grandma across a river. But there is no bridge!
Can YOU help her build a bridge?



What is a Bridge?

A bridge is a special path to cross water, roads, or big gaps safely.

Key Words: Bridge, Span, Gap, Support, Pillar, Load, Engineer

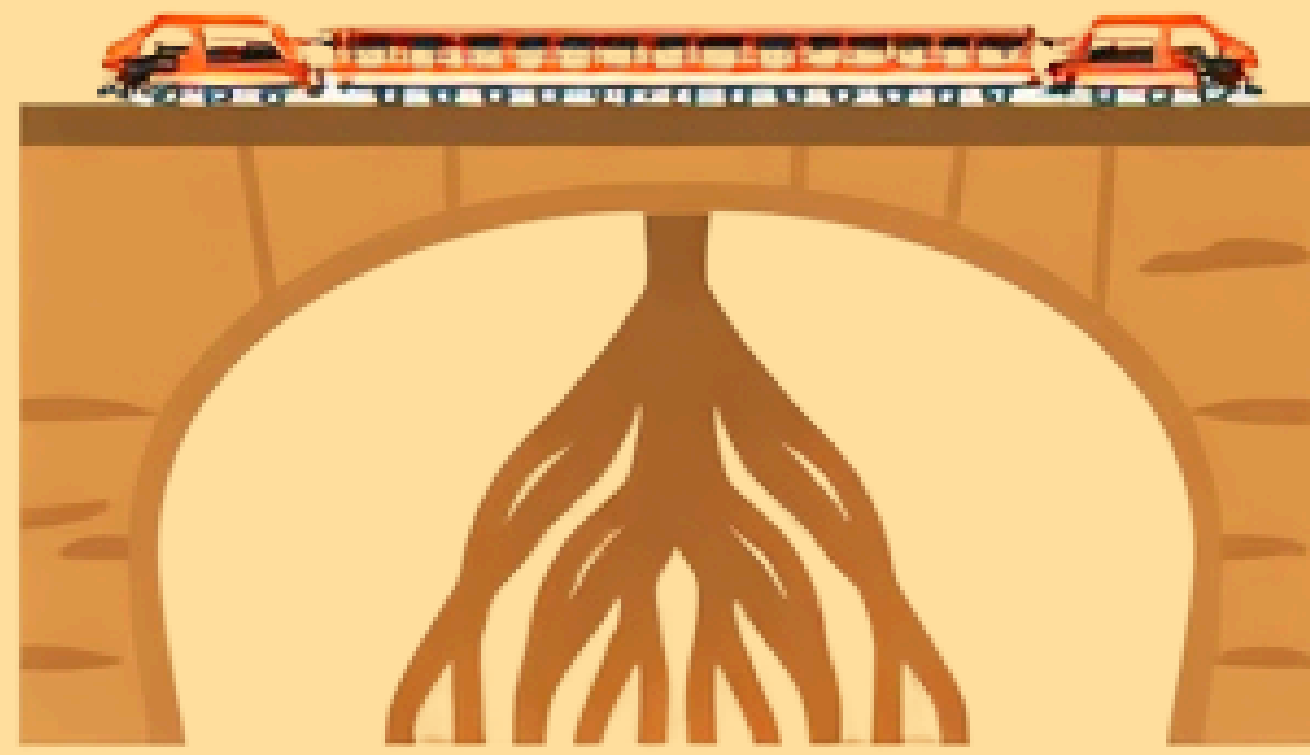


Bridges in India

🌉 Howrah Bridge: Thousands cross daily over Hooghly River, Kolkata

🚂 Pamban Bridge: Trains cross the sea in Tamil Nadu

🌿 Living Root Bridges: Made from tree roots in Northeast India





Why Do We Need Bridges?

- Connect hard-to-reach places
- Help children go to school during floods (Brahmaputra, Godavari rivers)
- City flyovers reduce traffic jams (Bengaluru, Chennai, Delhi)
- Railway bridges connect islands to mainland





Types of Bridges - Beam Bridge

A beam bridge is like a flat plank across a gap!

Examples: Wooden plank over a village stream, road flyovers

Try it: Put a ruler between two stacks of books - that's a beam bridge!



Types of Bridges - Arch Bridge

An arch bridge has a curved shape that makes it very strong!

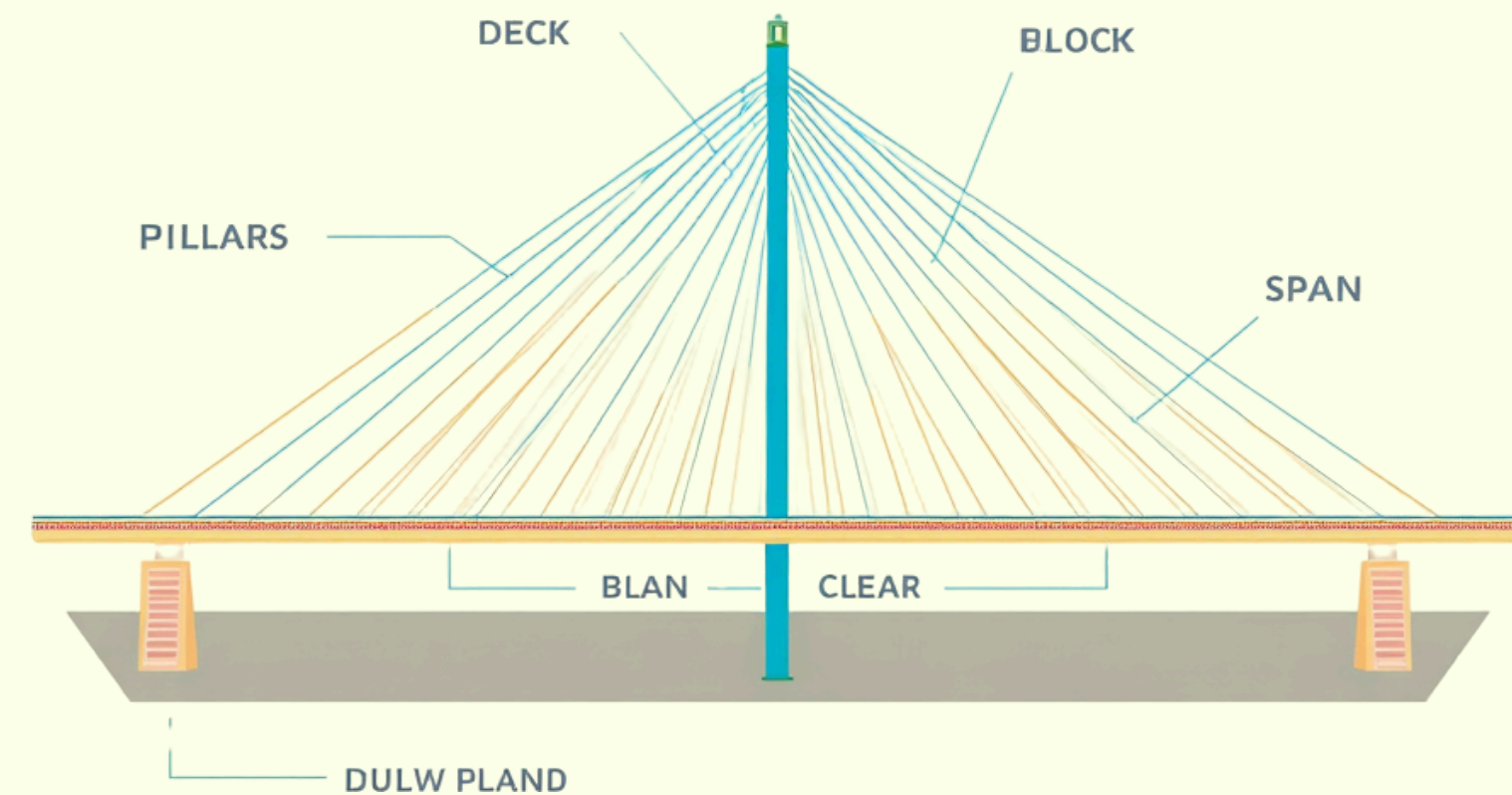
Examples: Howrah Bridge, old stone bridges in Rajasthan, Kerala, and Kashmir.

Hands-on: Make an arch with your hands and feel its strength!



Parts of a Bridge

- Deck: The flat part you walk or drive on
 - Support/Pillars: The legs that hold the bridge up
 - Span: The distance from one side to the other
- Example: Bandra-Worli Sea Link has tall pillars and a 5.6 km span!

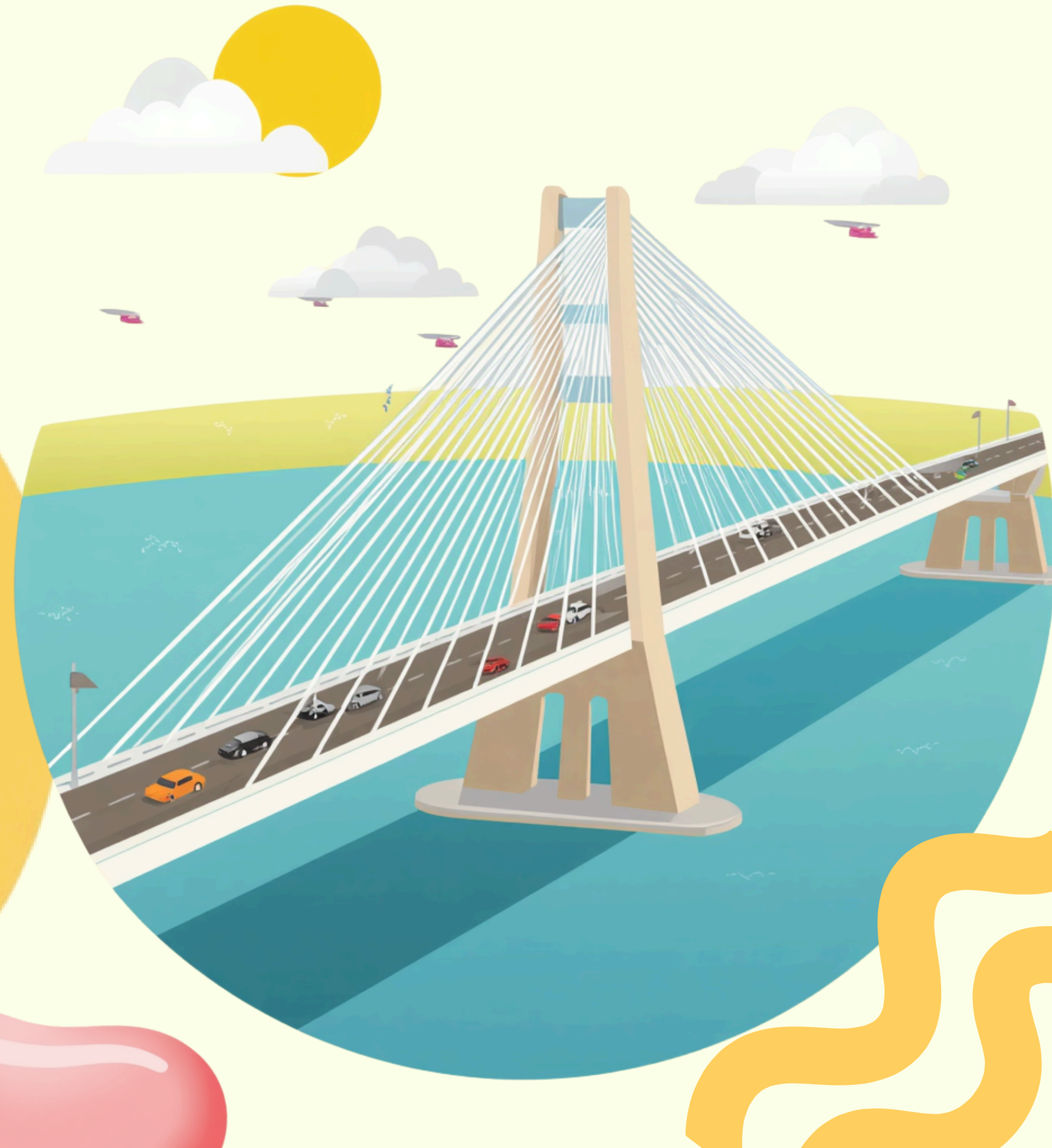


What Makes a Bridge Strong?

A bridge must hold its own weight AND everyone crossing it! Engineers test bridges before people use them.

🏛️ Activity Preview: We will test our bridge with coins!

🌟 Indian Spotlight: Atal Setu (Mumbai) is 21.8 km long and holds thousands of cars!





Build Activity - Plan Your Bridge

Step 1: Draw your bridge design!
Think about: Where will the pillars
go? How wide is the deck?
Materials you can use: books, paper,
coins, popsicle sticks, clay,
cardboard



Build Activity - Build & Test

Step 2 & 3: Gather materials and build your bridge across two stacks of books.

Step 4: Test by placing coins one by one in the center.

Count how many coins your bridge can hold!

Build Activity - Observe & Record

Step 5: What happened? Did the bridge hold, bend, or break?

Write or draw what you saw. Tell your friend about your bridge experiment!



Reflection Questions

Think about what you learned today!

- What is a bridge?
- Name one Indian bridge you learned about
 - What were the two types of bridges?
 - How many coins did your bridge hold?
- What would you change to make it stronger?



Quick Quiz Time!



A bridge helps us ____

a) Eat food b) Cross a gap c) Sleep

Howrah Bridge is in ____

a) Mumbai b) Kolkata c) Chennai

The flat part you walk on is called ____

a) Pillar b) Span c) Deck



Congratulations, Engineer!

Meera crosses her bridge! You built your FIRST
bridge today!

Great job, Engineer! Keep building and improving.

